TIANSAWANG SUETRONG

CURRICULUM VITAE | SOFTWARE ENGINEERING, BACKEND & DATA

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PROFILE

Digital Science and Technology student at the Faculty of ICT, Mahidol University, seeking a software engineering, backend, or data internship. I like owning the whole system: the interface people use, the API in the middle, and the data model underneath. I have built and deployed two full-stack systems that are live today: a no-code bot platform running on Rust WebAssembly, and a geospatial decision platform for Bangkok's urban heat that runs on real satellite and open data. I care about correctness: every number should be traceable to its source, and a system should say plainly when its data is missing rather than guess.

EDUCATION

Mahidol University | **2025 - Present**
Faculty of Information and Communication Technology (ICT)
Bachelor of Science in Digital Science and Technology | GPA 3.10
Relevant Coursework: Database Systems, Data Structures, Web Programming, Object-Oriented Programming

Klaeng Wittayasathaworn School | **2022 - 2025**
Secondary Education

TECHNICAL SKILLS

Programming: Python, TypeScript, JavaScript, Java, SQL
Web Development: HTML, CSS, Node.js, Express.js, Web Services
Databases: MySQL, Supabase
Data and Modeling: ER modeling, relational database design, data validation
Tools: Git, GitHub, VS Code, Linux, Postman, npm

PROJECTS

HEAT//SHIFT - Map-first urban heat decision platform for Bangkok
Live pilot, 2026 | GitHub Repository | Live Website
Python, FastAPI, Next.js, MapLibre, PostgreSQL/PostGIS, Redis/Celery, Google Earth Engine, OR-Tools, Docker, GitHub Actions
A platform that helps plan where to invest in cooling Bangkok, using real satellite, weather, population, and map data across all 50 districts.

Data pipeline

- Designed a six-stage pipeline: staging, validation, processing, publishing, cache invalidation, and live client updates over Server-Sent Events; invalid data fails the job and keeps the previous version.
- Connected seven real sources that update automatically: Bangkok district boundaries, Landsat 8/9 land surface temperature, Sentinel-2 NDVI, Dynamic World land cover, WorldPop population, Open-Meteo weather, and OpenStreetMap.
- Separates when a provider observed the data from when the system last checked it, and stores every published version so each map value can be traced to its dataset version.

Planning and analysis

- Built a budget optimizer with Google OR-Tools that compares strategies on candidate sites with referenced costs.
- Withholds forecasts that fail holdout validation and shows their errors, and keeps unverified sites out of the optimizer instead of inventing costs or benefits.
- Generated printable evidence reports with data versions, uncertainty, and stale-result warnings.

Operations and security

- Runs 24/7 on free-tier cloud hosting at $0 cost, with web, API, worker, and scheduler supervised inside one container under a 768 MiB memory limit.
- Protected admin tools with OpenID Connect single sign-on and role checks; production refuses demo data and shared keys.
- Added CI checks that build the hosted image and test it under the same CPU and memory limits as production.

WasmBot Studio - No-code business rule bot platform

Deployed, 2026 | *GitHub Repository* | *Live Website*

TypeScript, React, Vite, Tailwind CSS, Supabase (Auth, PostgreSQL, Edge Functions), Rust WebAssembly

A platform that lets non-developers turn decision rules into secure, shareable bots.

- Built a visual editor for nested AND/OR rules with 12 comparison operators, executed by a Rust rule engine compiled to WebAssembly rather than simulated in the browser.
- Designed immutable, versioned deployments so draft edits never affect the live version.
- Exposed each bot through a hosted form, iframe/script embed, REST API, signed webhooks, and a typed TypeScript SDK.
- Secured data with Supabase Auth, PostgreSQL row-level security, scoped API keys stored as HMAC-SHA256 hashes, rate limits, domain allowlists, and audit logs.
- Built webhook delivery with AES-GCM encrypted secrets and a transactional outbox with up to three attempts.
- Integrated the WASM engine with seven Supabase Edge Functions and verified the system with 71 automated tests, including production, authentication, authorization, and security-header checks.

Other work

- **Personal portfolio** (website): buildless, responsive site in HTML, CSS, and JavaScript with dark mode, keyboard navigation, reduced-motion support, and a print layout.
- **HTML101** (GitHub): web fundamentals labs covering semantic HTML, layout, styling, and multi-page sites.

EXPERIENCE AND ACTIVITIES

MUICT Open House - Guest Relations | Mahidol University | 2026
Raknong G2 - Staff | Mahidol University | 2026

LANGUAGES

Thai: Native | English: Intermediate

INTERESTS

Web and software: designing systems that fit together, from the database schema to the button people click.
Data and geospatial: turning messy real-world data into versioned, trustworthy decisions on a map.
AI and networks: exploring intelligent applications and how devices communicate.

REFERENCES

Available upon request.